

Milestone 2 Summary Document

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CS 4412 Data Mining

Key EDA Insights:

Typically play count follows an exponential distribution, most maps have a low playcount, while a few "mega-hits" garner millions of plays. By applying a logarithmic transformation, we can better visualize the spread of the data.

There is a clear difference between the BPM of the top 10% of songs and the rest. Lower BPM maps, generally in the 75-150 range, are significantly less represented in the top 10% of maps compared to higher BPM maps. There is a clear spike around the 175 BPM range where a large majority of maps of maps in this range are in the top 10% of playcount. This indicates that players prefer songs with a higher bpm and that it directly contributes to the popularity of a map.

Often, maps that are too easy or too overwhelmingly difficult have lower playcounts, while those in the mid-to-high difficulty range (e.g., 4 to 6 stars) attract the largest player base because they are challenging but accessible.

The heatmap showed the correlation between different numerical features. There is a clear correlation between playcount and difficulty rating. This shows that the more difficult a map is, generally it receives more plays than easier maps. Another example is the interesting correlation between a songs rating and its max combo, hinting that songs with a higher max combo generally receive a better rating than maps with a lower max combo.

Preprocessing and Decision Rational:

Before EDA, Both columns and rows were adjusted to remove columns that aren't relevant and to remove rows that aren't relevant. Specifically, this includes: > Remove all rows where mode is not 0 (the standard osu game mode).

Remove all rows that are either pending (0), WIP (-1), or graveyard (-2)

Remove all rows where the total song time is less than 40 seconds or greater than 2 standard deviations from the mean.

Remove all rows where the Drain Ratio (Hit length / Total Length) is less than 0.6, This means that the map is less than 60% game time and a large part is empty audio (intros, outros, massive breaks) and is a good indicator of > a low-quality beatmap.

Remove maps where there are impossible statistics / data collection inaccuracy (bpm $\geq$ 500, difficulty rating $\geq$ 11)

Before applying the k-means mining technique, the highly skewed variables (replay ratio and favorite count) had to be logarithmically skewed to ensure they don't overpower the algorithm.

First Mining Results:

The k-means graph with $k=4$ showed that there were distinct groups of maps, even if they may overlap a bit. It also showed how stats like BPM and difficulty rating are very important in determining the popularity of a map and its favorite count, as was originally found in the heatmap from the EDA section.

How the results relate to the discovery questions:

Question 1: What map and song characteristics are most commonly shared among the most popular beatmaps?

By clustering, we can see that "popular" maps (high favorite and decent plays) do not all look the same. Cluster 2 reveals a highly popular segment driven by high BPM (~180+) and a high difficulty (~5). In contrast cluster 0 achieves popularity through shorter, easier gameplay (primarily targeting newer players).

Question 2: Are there distinct groups of songs that achieve popularity in different ways (high replay vs high appreciation)?

Cluster 2 (The Grind), perfectly defines the high replay category. The replay ratio (2.25) is massively higher than any other cluster. These maps have high difficulty, but they still achieve lots of plays from players constantly failing and retrying them, inflating their playcount relative to their passcount.

Cluster 0 (Experiences), represents the maps within the high appreciation category of maps. They have very low replay ratios (meaning players often pass them on the first few tries) but maintain incredibly high favorite counts. Players play them for the music, the storyboard, or the mapping quality, not to grind performance points or difficulty.

Milestone 3 Changes: Currently nothing in milestone 3 needs to be changed.